Rhizosphere 22 (2022) 100510

Contents lists available at ScienceDirect

Rhizosphere

journal homepage: www.elsevier.com/locate/rhisph

Influence of living grass Roots and endophytic fungal hyphae on soil hydraulic properties

Katelyn M. Marcacci[a], Jeffrey M. Warren[b,*], Edmund Perfect[a], Jesse L. Labbé[c,d]

[a] Department of Earth and Planetary Sciences, University of Tennessee – Knoxville, Knoxville, TN, 37996, USA
[b] Environmental Sciences Division and Climate Change Science Institute, Oak Ridge National Laboratory, Oak Ridge, TN, 37831, USA
[c] Biosciences Division, Oak Ridge National Laboratory, Oak Ridge, TN, 37831, USA
[d] Invaio Sciences Cambridge, MA, 02138, USA

ARTICLE INFO

Keywords:
Switchgrass (*Panicum virgatum*)
Serendipita indica
Porosity
Soil water retention
Hydraulic conductivity

ABSTRACT

Soil hydraulic properties are often estimated based on laboratory data or pedotransfer functions dependent on soil physical properties, which often do not consider potential impacts of soil roots or fungal hyphae. Here, we first review current knowledge of how these soil biotic components affect hydraulic properties, then we conducted laboratory experiments to specifically test if the presence of roots and mycorrhizal fungi had a significant effect on the hydraulic properties of two soils with contrasting textures: Flint sand and Hamblen silt loam. Soil cores were seeded with (*Panicum virgatum*) and grown in a greenhouse over three separate growth periods. The endophytic fungus *Serendipita indica* was injected as liquid inoculant into designated mycorrhizal cores. Saturated hydraulic conductivity (K_{sat}) measurements were made with a constant head permeameter, and soil water retention curves were obtained by the evaporation method, supplemented at the dry end for Hamblen silt loam with water activity meter data. Retention curve parameters were obtained by fitting the van Genuchten equation to the resulting measurements. Mean root volume ratios were higher in the mycorrhizal inoculated treatment than in the uninoculated treatment for both soils. For Flint sand, analysis of variance revealed that K_{sat} was reduced by the presence of roots as compared to bare soil. This was likely due to roots clogging soil pores. Results also indicated the presence of roots changed the shape of the water retention curve for Flint sand by increasing water content at saturation and by reducing the slope of the curve. These changes suggested roots created additional porosity and broadened the pore-size distribution. The presence of mycorrhizal fungi accentuated the root effects. The influence of roots and mycorrhizal fungi on hydraulic properties was less obvious for the Hamblen silt loam, as none of the treatments differed from each other at $p < 0.05$. The results highlight the necessity to consider the impact of root and fungal structures on models of soil hydraulic properties.

1. Introduction

Although soil moisture is a fundamental land surface variable, its influence on climate is difficult to quantify (Robinson et al., 2019). There is a need for improved characterization and validation of soil water-vegetation relationships (Seneviratne et al., 2010). The inability of many models to account for variations in water content and availability due to soil structure contributes to uncertainty in their predictions (Jarvis et al., 2013; Fatichi et al., 2020). Also, the biological feedback between soil and vegetation makes it challenging to incorporate meaningful hydraulic parameters into climate models (Robinson et al., 2019).

Roots play an essential role in the water balance, where soil-plant hydrologic feedback is dependent upon water availability and depth of infiltration (Oswald et al., 2008). Roots alter soil hydraulic properties by changing the pore size distribution, with vegetated soil favoring the creation of macropores (Rachman et al., 2004). Changes in pore space geometry, when compared to bulk soil, are due to: (i) the presence of roots in soil pores, (ii) root water retention, and (iii) root exudates (Leung et al., 2015a). Impacts vary with the age and physical characteristics of the root system; e.g., younger, finer roots have greater water uptake rates per surface area than more mature roots, impacting near root soil hydration dynamics (Dhiman et al., 2018). Indeed, the soil region directly surrounding roots, the rhizosphere, has quite different hydraulic properties than the surrounding bulk soil (Carminati et al., 2010; Querejeta, 2017). The water holding capacity of this region is governed by pore-size distribution, the chemical composition of root

* Corresponding author.
E-mail address: warrenjm@ornl.gov (J.M. Warren).

https://doi.org/10.1016/j.rhisph.2022.100510
Received 13 December 2021; Received in revised form 21 March 2022; Accepted 29 March 2022
Available online 5 April 2022

Abbreviations

ANOVA	Analysis of Variance
CON	Control
FS	Flint Sand
FRT	Fertilizer
HSL	Hamblen Silt Loam
MYC	Mycorrhizal
RMSE	Root Mean Square Error
RTS	Rooted

exudates (or mucilage) or lysed cells, and its previous wetting and drying history (Czarnes et al., 2000). The water content of the rhizosphere is generally higher than that of bulk soil due to the enhanced water holding capacity of root mucilage (Carminati, 2012; Ahmed et al., 2016; Dhiman et al., 2018). There is also evidence that changes in the water content of the rhizosphere can be inversely related to those occurring in the bulk soil due to hysteresis dynamics (Carminati et al., 2010).

Mycorrhizal fungi play a significant ecological role in soils, but little is known about the quantitative effects they have on soil hydraulic relations and scaled impacts on climate (Brito et al., 2009). Furthermore, it can be difficult to separate out the impact of fungal hyphae on hydraulic properties from that due to roots alone because of the complexity of their symbiotic relationship (Gehring, 2017). Mycorrhizal hyphae provide water and nutrients (especially phosphorus) to the plant in return for root assimilates (Van Der Heijden et al., 2006). They also improve rhizosphere root-hydraulic properties by redistributing water and nutrients from bulk soil to roots along preferential flow paths created by fungal hyphae (Augé et al., 2001; Warren et al., 2008). Mycorrhizal fungi are known to change soil hydraulic properties by enmeshing soil particles with hyphae, promoting the formation of aggregates, and increasing aggregate stability. Soil aggregate formation increases the creation of macropores, thereby changing the pore-size distribution and reducing bulk density. Fungal hyphae exudates can increase soil water retention leading to more water availability in soils during drying and increased transmissive capacity (Augé et al., 2001; Querejeta, 2017).

1.1. Roots and mycorrhizal effects on porosity

A growing root can radially deform the surrounding soil by cylindrical expansion. Dexter (1987) developed a simplified model for this compaction process based on the assumption that the volume occupied by the root is accommodated by the loss of an equal volume of pore space in the rhizosphere. Localized root-induced compaction within the rhizosphere has been imaged using x-ray computed tomography (Aravena et al., 2011); Daly et al., 2015) and the extent of root-induced compaction depends upon the particular plant species investigated (Whalley et al., 2005). Further, root biomass density is inversely related to soil porosity (φ) (Jotisankasa and Sirirattanachat, 2017).

The effects of mycorrhizal hyphae on φ have also been investigated in several studies involving controlled pot experiments. Results vary due to the type of soil or plant – fungal association used; the former varying in soil physical and hydraulic properties, the latter varying in biotic traits such as root and hyphal length, diameter, distribution and release of exudates. A number of studies have shown increased soil porosity in the presence of mycorrhizal hyphae compared to uninoculated controls, including in onion (*Allium cepa* L.) inoculated or not with *Glomus macrocarpum* (Thomas et al., 1986), *Medicago polymorpha* L. (Akhzari, 2015) and barley (*Hordeum vulgare* L.; Samaei et al., 2015). In contrast, Daynes et al. (2013) found no effect of mycorrhizal hyphae on φ in a study involving one grass and two woody species growing in compost applied to coal mine spoil. More recently, φ was significantly *reduced* in a tomato (*Solanum lycopersicum*) - *Rhizoglomus irregulare* system growing in natural sandy soil, fine sand or vermiculite (Bitterlich et al., 2018).

1.2. Root and mycorrhizal effects on soil water retention

The soil water retention curve describes the relationship between soil water potential a soil water content, $\theta(h)$, and is commonly parameterized by fitting a mathematical function to experimental data (Dane and Topp, 2002). Numerous functions have been used to fit $\theta(h)$ data; those that have been employed in plant root and/or fungal mycorrhizal studies include the van Genuchten model (1980), the Kosugi (1996) model, the Gallipoli et al. (2003) model, the Daynes et al. (2013) model, the Augé et al. (2001, 2004) model, and the Durner (1994) model. Most of these models contain the saturated (θ_s) and residual (θ_r) volumetric water contents, plus various fitting parameters related in some way to a benchmark capillary pressure head and the breadth of the pore-size distribution (Table 1).

Several studies have investigated the influence of roots on θ_s. Daly et al. (2015) found that θ_s was higher in non-rhizosphere soil than in the rhizosphere. Conversely, Powis et al. (2003), Carminati et al. (2010) and Leung et al. (2015b) suggested that soil with roots has a higher θ_s than unrooted soil. Root-dense soil had greater water retention and θ_s, but lower saturated permeability compared to less dense root layers in bio-engineered field slopes (Rajamanthri et al., 2021). Bodner et al. (2014) found that θ_s had a positive logistical relationship to root density, while Shao et al. (2017) accounted for a higher θ_s in densely-rooted soil when formulating a dual-permeability model. Scanlan (2009) found that although wheat roots had a radius large enough to influence porosity, there was no impact on θ_s. These trends suggest that the influence of roots is species and time dependent, with root density and compaction being key factors in influencing θ_s.

The effects of roots on θ_r have proven to be either inconsistent or unreported. Yan and Zhang (2015) found that lower density soils, generally associated with vegetation, desaturated more than bare soils. Daly et al. (2015) found that root-permeated soil drained to a lower θ_r than bare soil, but their imaging model for x-ray tomography greatly overestimated θ_r. Other studies have shown θ_r be consistent across vegetated and bare soil (Leung et al., 2015b), or even higher in vegetated soil due to the influence of root-exuded mucilage Carminati et al. (2010); Ng et al. (2016).

Ng et al. (2016) have proposed a simple model to account for the effect of roots on the $\theta(h)$, based on a root-dependent void ratio term to account for the change in root volume ratio in root-permeated soil, *i.e.,*

$$e_r = \frac{e - R_v\,(1+e)}{1 + R_v\,(1+e)} \quad [1]$$

where e_r is the void ratio of root-permeated soil, e represents the void

Table 1
Capillary pressure head and pore-size distribution parameters for various mathematical water retention models.

Mathematical model	Capillary pressure head parameters	Pore-size distribution parameters
van Genuchten (1980)	α	n, m
Kosugi (1996)	h_m	σ
Gallipoli et al. (2003)	ψ, ω	n, m
Daynes et al. (2013)	χ	β
Augé et al. (2001, 2004)	λ	μ, δ
Durner (1994)	α	n

ratio of the same soil without roots, and R_v is the total volume of roots per unit volume of soil. Application of this void ratio into a $\theta(h)$ model to account for root-induced changes in water retention indicates that root occupation of soil pores could alter the ψ and ω capillary pressure head parameters (Ng et al., 2016).

Scholl et al. (2014) found that h_m, corresponding to the pressure head at 50% effective saturation, was not statistically different for $\theta(h)$'s from unrooted versus rooted soil. In contrast, root density has been shown to influence the α parameter, which accounts for the vertical break in the $\theta(h)$ at air entry, through time, although results are mixed. Shrub planting density in a silty sand soil resulted in higher α values in one study (Shao et al., 2017), while others found α decreased with root density (Leung et al., 2015b); Jotisankasa and Sirirattanachat (2017) Within the near root rhizosphere, α was shown to be lower than in non-rhizosphere soil (Carminati et al., 2010), indicating spatial differences in α within bulk soil.

Roots also influence the slope of the $\theta(h)$ which is characterized by the pore-size distribution parameters in Table 1. Yan and Zhang (2015) found that vegetation did not influence the slope of the $\theta(h)$ in high permeability sandy soil but increased slope steepness in cemented low permeability soil. This result is supported by Shao et al. (2017), who found that the slope became steeper with planting density. Daly et al. (2015) found that, although $\theta(h)$'s derived from x-ray tomography data poorly estimated n, they were consistent in having higher n values and steeper slopes in root-permeated soil than in soil without roots, although the opposite pattern was reported by Leung et al. (2015b) The σ parameter has also been found to be higher in root-permeated soil compared to bare soil (Scholl et al., 2014). Rachman et al. (2004) observed that the $\theta(h)$ became less steep as plant roots spread, and bulk density decreased. Carminati et al. (2010) found that n was lower, and $\theta(h)$ was less steep in root-permeated soil than in bulk soil, and attributed this to the water retention capacity of root mucilage.

There is limited literature available comparing $\theta(h)$ parameters between mycorrhizal and non-mycorrhizal samples. Hosseini et al. (2016) found that α and n were both lower in endophyte-fungi inoculated soil when compared to non-inoculated soil. Similarly, while statistically comparing parameters derived from the Durner (1994) model, Bitterlich et al. (2018) found that $\theta(h)$ curves for mycorrhizal-inoculated samples had increased slopes and shifted capillary pressure benchmarks compared to the controls. Augé et al. (2001, 2004) found that soil containing roots and mycorrhizal fungi initially had more water loss than rooted, non-mycorrhizal soils, and more water was available in the root + mycorrhizal soil. Daynes et al. (2013) found that β and the entire $\theta(h)$ for fungal-inoculated samples was slightly, but not significantly higher than the non-inoculated samples.

1.3. Root and mycorrhizal effects on saturated flow

In root-permeated soil, the flow of water under fully-saturated conditions is largely influenced by root density. Pore blockage by roots has been shown to reduce the infiltration rate (e.g., Barley, 1954) and reduce the saturated hydraulic conductivity (e.g., K_{sat}; Leung et al., 2015a) and the reduction in K_{sat} is correlated to root biomass (Jotisankasa and Siriattanachat, 2017). Shao et al. (2017) found that K_{sat} is dependent upon planting density, with a low root density decreasing K_{sat} (in agreement with the other studies) but a 10X higher root density increasing K_{sat}, potentially due to more dead roots impacting soil hydraulic characteristics.

The relationship between K_{sat} and roots appears to be age dependent. Young, active roots can excrete mucilage and exudates and have greater rates of water and ion uptake than older, larger or suberized roots (Czarnes et al., 2000; Ahmed et al., 2016; Dhiman et al., 2018), which can all impact soil hydraulic properties. Scanlan (2009) determined there was a noticeable, but not statistically significant, increase in K_{sat} during the changeover from vegetative to reproductive growth in 7-9-week-old wheat plants. Petersson et al. (1987) found a positive linear relationship between age and K_{sat} for older roots, while younger roots had a less direct, and negative relationship between increased root development and K_{sat}. These trends were supported by the work of Powis et al. (2003), who showed that K_{sat} decreased during root growth, but increased during root decay. Ni et al. (2018) modeled the relationship between root growth and decay in soil void space. The decay of structural roots from older plants increased macroporosity and thereby K_{sat}, in agreement with the field observations of Shao et al. (2017). Scholl et al. (2014) estimated K_{sat} inversely from cumulative outflow data and found an increase over time in planted soil columns as compared to unplanted columns.

Rachman et al. (2004) found that a high incidence of inter-aggregate pore spaces (mesopores) corresponded to higher K_{sat} values in switchgrass-planted soil, but the presence of mycorrhizal fungi decreased K_{sat}. Similarly, Samaei et al. (2015) found that mycorrhizal fungi increased soil aggregation, mesopores and micropores and decreased K_{sat} when compared to non-mycorrhizal plants. Likewise, Bitterlich et al. (2018) found that K_{sat} was lower in inoculated samples than in non-mycorrhizal samples. These differences in K_{sat} can be attributed to mycorrhizal derived changes in pore-size distribution, as well as the presence of significantly more water-stable aggregates in soils with mycorrhizal-plant symbiosis (Hallett et al., 2009).

1.4. Root and mycorrhizal effects on unsaturated flow

In unsaturated soil, the hydraulic conductivity depends upon the volumetric water content, yielding a functional relationship, $K(\theta)$. The $K(\theta)$ curve is greatly influenced by pore space geometry. During drainage, the largest pores drain first. Since these conduct the greatest amount of water, $K(\theta)$ can decrease by several orders of magnitude with just a small decrease in water content below θ_s. Nearly all soil-water interactions, including the supply of water to roots, take place in unsaturated soils (Hillel, 2004). However, relatively few studies have investigated the effects of plant roots on $K(\theta)$.

When testing plants grown in a synthetic sand-clay mixture, Powis et al. (2003), did not observe any relationship between root volume or root length and $K(\theta)$. In another study, root compaction of clay soil reduced void spaces and increased contact between aggregates, resulting in higher unsaturated $K(\theta)$ under partially saturated conditions when compared to unrooted control soil aggregates (Aravena et al., 2011). Sedgley and Barley (1958); however, observed that a sandy loam planted with grass had lower $K(\theta)$ compared to control samples. The reduction in $K(\theta)$ was theorized to be the result of pore-clogging by fine roots and soil compaction by larger roots as well as hydrophobic root exudates. Macropores created by root decay were not considered an influencing factor because these samples were tested under conditions where larger pores would have already drained. The contradicting results between Aravena et al. (2011), and Sedgley and Barley (1958), could be related to the differences in soil type investigated. Unsaturated flow tends to be higher in clay soils than in coarse sands, suggesting that roots increase $K(\theta)$ under high flow conditions, and decrease them under low flow conditions. Further studies are needed to fully investigate the full range of root impacts on unsaturated flow.

When compared to non-mycorrhizal soils, soils inoculated with mycorrhizal fungi have been shown to have greater pore connectivity and flow during unsaturated conditions (Querejeta, 2017). Mycorrhizal fungi causes $K(\theta)$ to decrease with decreasing water content at a lower rate compared to uninoculated soil. This effect is believed to be caused by a combination of three main factors. Firstly, mycorrhizal fungi change the shape and organization of soil pores towards smaller pore spaces (Samaei et al., 2015) and $K(\theta)$ is known to decrease more gradually in micropore dominated soil (Scott, 2000). Secondly, mycorrhizal fungi influence $K(\theta)$ by improving pore connectivity. Preferential flow through hyphal networks allows for greater unsaturated flow when compared to non-mycorrhizal soils (Querejeta, 2017). Thirdly, mycorrhizal fungi influence $K(\theta)$ through the exudation of organic compounds

(Bitterlich et al., 2018). These compounds are strongly hydrophilic and increase conductivity within water-filled pore spaces when compared to non-mycorrhizal soils (Hallett et al., 2009). The combination of these three factors can result in a "truly mycorrhizal" effect on $K(\theta)$ (Bitterlich et al., 2018).

1.5. Goal, objectives, and hypotheses

While we have highlighted prior studies of root impacts on soil hydraulic properties, and much less conducted on mycorrhizae, more research across soil types and with different plant-fungal associations is needed to improve our predictive capacity of soil-root-fungal hydrological interactions and their linkages to solute and nutrient exchanges. Broader issues related to water infiltration, storage, runoff and slope stability are also intricately linked to root distribution, traits and impacts on soil properties. With potential changes in precipitation and drought frequency and intensity expected due to climate change, such research may be particularly important to understand the impact of novel hydrological regimes on different ecosystems.

Research into the unique hydraulic properties of vegetated soil can thus provide a better understanding of the relationship between soil water and climatic conditions. Key soil hydraulic properties include porosity, the water retention curve and saturated and unsaturated hydraulic conductivities. This paper seeks to investigate the sensitivity of these properties, for possible inclusion in the hydrological module of predictive climate models, by measuring them in the presence and absence of living roots and endophytic fungal hyphae. Our main objective was to experimentally investigate in the laboratory, how the presence and absence of mycorrhizal fungi, in conjunction with plant roots, impact soil hydraulic properties. We further will quantify total plant biomass and root volume in the presence and absence of fungal hyphae in order to seek statistical relationships between these biological parameters and soil hydraulic properties. While we have shown various and sometimes contradictory prior results of roots and fungi on soil hydraulic properties, here we hypothesize that:

1. Roots and fungal hyphae partially fill macropores, decreasing the saturated hydraulic conductivity (K_{sat}) of the soil;
2. Roots and mycorrhizal fungi increase the volumetric water content at saturation (θ_s) reducing the slope of the soil water retention curve $[\theta(h)]$, with a larger effect for coarser textured soil;
3. Roots and mycorrhizal fungi increase pore connectivity during drying, increasing the unsaturated hydraulic conductivity $[K(\theta)]$ of the soil, with a smaller effect for coarser textured soil;
4. Roots alone will have less impact on soil hydraulic properties than roots inoculated with mycorrhizal fungi.

2. Materials and methods

2.1. Experimental design

Bench-scale measurements of water retention and saturated and unsaturated hydraulic conductivity were performed in a laboratory setting using a total of 42 hand-packed soil cores. Two different soil types were used: Flint sand (Flint #13, U.S. Silica Company, Berkeley Springs, WV) and Hamblen silt loam. Selected physical and chemical properties of the two soils are summarized in Table 2.

Flint sand was selected because it is a homogenous, non-biological porous medium, with predominately quartz composition, negligible organic matter, high saturated hydraulic conductivity, narrow pore-size distribution, and rapid drainage time (Kang et al., 2013). Hamblen silt loam was selected to represent a naturally-occurring soil type within East Tennessee. The soil is formed in watersheds of eroded limestone, sandstone, and shale, and is considered ideal for agriculture (NRCS, 2019). Soil samples were obtained from the B horizon, air-dried, and passed through a 2-mm sieve.

Table 2
Selected physical and chemical soil properties of Flint sand (FS) and Hamblen silt loam (HSL).

Soil	Texture	Organic Matter (*wt* %)	pH	Specific Gravity	CEC (*cmolc/kg*)
FS	Sand	0[d]	6.0–8.0[a]	2.65[a]	0.6[b]
HSL	Silty clay loam	0.2–1.0[c]	5.0–7.0[c]	2.65[d]	3.7–7.6[c]

[a] US Silica Company (2010)
[b] Zhuang et al., (2003).
[c] Soil Survey of Roane County Tennessee (NRCS, 2019).
[d] Assumed Value.

The plant associate used in the experiments was *Panicum virgatum* "colony type" switchgrass, and the mycorrhizal fungal inoculant was *Serendipita indica* (*Piriformospora indica*, ATCC 204458) (Verma et al., 1998; Weiβ et al., 2016). Switchgrass and *Serendipita* spp. make an excellent experimental system which facilitates a reproducible setting in bioenergy crop studies (Ray et al., 2021). *S. indica* can be propagated axenically such that it can exist in soil with or without a host plant, which allows independent assessment of the effect of the fungi versus roots on soil properties. In addition, switchgrass is one of the plant models suited to marginal cropland which is an energetically and economically feasible and sustainable biomass energy crop with currently available technology. It has been identified as an important bioenergy crop for cellulosic ethanol production in the United States (Adams and Kszos, 2005).

For the Flint sand, there were five treatments: control (CON), rooted (RTS), fertilizer rooted (FRT), mycorrhizal rooted (RTS + MYC), and mycorrhizal control (CON + MYC), each replicated three times. The FRT treatment consisted of a one-time application of 15-9-12 N–P–K fertilizer plus micronutrients at 2 g L^{-1}. The fertilizer solution was applied after the seeds had germinated and started growing. Due to equipment limitations and logistics, two separate 70-day growth cycles were investigated to increase replication (i.e., a total of 30 cores; Table S1).

For the Hamblen silt loam soil, there were four treatments: control (CON), rooted (RTS), mycorrhizal rooted (RTS + MYC), and mycorrhizal control (CON + MYC), each replicated three times. A fertilizer treatment was not included due to the presence of naturally-occurring nutrients in this soil. Additionally, only one 80-day growth cycle was investigated because of the slow drainage and drying during the soil water retention curve measurements (i.e., a total of 12 cores).

2.2. Soil preparation

Both soils were autoclaved at 111.5 kPa and 121 °C for 30 min. After 24 h they were autoclaved again to ensure the neutralization of resilient fungal spores (Brito et al., 2009). The autoclaved soils were subsequently oven-dried at 105 °C for 24 h to remove any moisture before packing into sample containers.

The sample containers were stainless steel and cylindrical in shape with an inner diameter of 4 cm and a height of 5 cm (Meter Co, Munich, Germany). The base of each cylinder was fitted with 1-μm nylon mesh (ELKO Filtering Co, Miami, Florida) to contain the soil and prevent exterior root and hyphal growth. As with the soil, the sample containers were autoclaved twice at 121 °C for 30 min to further neutralize any fungal spores (Brito et al., 2009). The sterilized oven-dried soil was weighed into equal parts and packed into the sterilized sample containers by hand to the same bulk density across replicates. Soil cores that were not selected for mycorrhizal inoculation were then wetted with DI water using a standing water table.

2.3. Mycorrhizal inoculation

Serendipita indica (*Piriformospora indica*, ATCC 204458) (Verma et al., 1998; Weiβ et al., 2016) was grown from a 0.5-cm agar plug in 50 ml of 1× potato dextrose broth (Sigma Aldrich, USA) in the dark at 30 °C (120 rpm shaker) for two weeks. These liquid cultures were then filtered using sterile Miracloth and rinsed twice with sterile water. Following autoclaving, the stainless-steel cylinders were UV sterilized for 30 min then packed with the autoclaved soil. The packed soil cylinders were then saturated with sterile DI water from below and 1 ml of *S. indica* liquid inoculant containing ~10^5 spores per ml was injected into the moistened soil to create a broad matrix of fungi.

2.4. Switchgrass seeding, germination, and growth

Inoculated and uninoculated soil cores were seeded with 3/8 teaspoon of *Panicum virgatum* "colony type" switchgrass seeds. The seeds were sprinkled uniformly on top of the soil surface. Immediately after application, and periodically, the seeds were misted with DI water to promote germination. After seeding and/or inoculation, all of the soil cores, including the controls, were placed in a greenhouse. The cores were watered daily with DI water by maintaining a standing water table and through spray bottle misting. The standing water table was gradually lowered as the plants matured. Seedlings were counted 4 weeks after application of seeds to the soil cores indicating initial germination rates of 1.75 seedlings per cm^2; more seeds germinated over time. Spore germination and presence of hyphae were confirmed in the controls by trypan blue staining of soil specimens with bright light optical microscopic observation, e.g., Fig. 1. While we did not quantify the distribution of mycorrhizal fungi in the soil, future work could assess this by various approaches, including trapping mycelium within the root system (Müller et al., 2017) and molecular characterization and sequencing approach of the ITS/LSU gene regions as a marker for general fungi and SSU region for arbuscular mycorrhizal fungi (Moll et al., 2016).

Based on Guha et al. (2018), the average day- and night-time air temperatures in the greenhouse were 24 and 18 °C, respectively; the average day and night-time relative humidity was 35 and 41%, respectively; and photosynthetically active radiation was 500 μmol m^{-2} s^{-1}. The plants grown in the Flint sand soil were harvested after 70 days, and the plants grown in the Hamblen silt loam were harvested after 80 days.

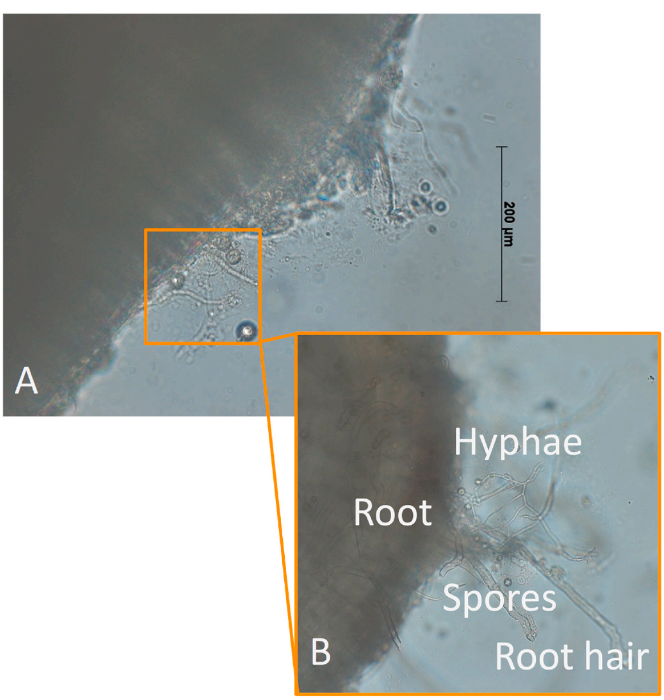

Fig. 1. The presence of fungal hyphae in the endophytic *Serendipita indica* inoculated treatments was confirmed using microscopy: (A) was taken under 20× magnification and shows a switchgrass root with mycorrhizal hyphae and fungal spores. (B) closeup of a similar area at 40× magnification. Relative diameters of structures: typical root (~200 μm), root hair (~12 μm), fungal hyphae (~3.5 μm).

2.5. Switchgrass biomass

The above ground shoot biomass was removed from the cores prior to measuring the soil hydraulic properties. It was oven-dried for three days at 60 °C and then weighed. Below ground root biomass was separated from the soil medium through root flotation in DI water after the soil hydraulic measurements were completed. The roots were gently picked from the soil, untangled, and the root nodes were removed. Some roots were set aside for root scanning and analysis (see section 3.6), then all roots were oven-dried for three days at 60 °C. Total biomass was calculated as the sum of the oven dry weight from the above ground and below ground biomass measurements.

2.6. Switchgrass root volume ratio

The root volume ratio (R_v) is the ratio of total fresh root volume to the total volume of soil, corresponding to the volume of the cylindrical soil cores. The total root volume was determined by scanning of fresh roots. A sub-sample of extracted roots was distributed uniformly on a scanning tray while ensuring there was no overlap of root material or extension beyond the dimensions of the scanner. The trays were then scanned, and the resulting images analyzed for root volume using root scanning software (WinRHIZO; Regent Instruments Inc., Québec, Canada). The total root volume present within one soil core was determined by scanning the entire root system, which required 25 sub-samples. Total root volume was also estimated from the individual subsamples through a ratio comparing the weight of the scanned roots to the weight of the unscanned roots. The number of sub-samples needed to estimate the mean total root volume within a core was then determined statistically. It was found that four sub-samples estimated the true value with 95% confidence. As a result, the remaining cores were analyzed using four sub-sample scans per core, with total root volume estimated from the weight ratio of the scanned to unscanned roots.

2.7. Saturated hydraulic conductivity

Following harvesting of the above-ground biomass, the soil cores were fully saturated from below with a standing water table and then fitted into a permeameter (KSAT; Meter Co, Munich, Germany). This device uses a Mariotte bottle system to provide a constant head and collects real-time flow data through a software application. The flow data were analyzed with Darcy's law to obtain the saturated hydraulic conductivity (K_{sat}) of the soil.

2.8. Soil water retention curve

Following the K_{sat} measurements, the saturated soil cores were transferred to a Hyprop2 device (Meter Co, Munich, Germany) for soil water retention measurements. First, two boreholes (0.5 cm in diameter and 1.25 cm and 3.75 cm deep), were made in the base of each soil core. For the Hamblen silt loam, the soil extracted from the boreholes was set aside for additional water retention measurements using a water activity meter (see below). Two mini tensiometers on the Hyprop2 device were pushed into the boreholes. The combined device and soil core was then vertically rotated, to expose the open end of the soil to the atmosphere and placed on an electronic balance.

Capillary pressure head (h) measurements were collected as moisture evaporated from the soil surface, and water was drawn out of the tensiometers. These data were recorded as $\log_{10}(h)$ in pF units. Changes in the weight of the soil cores were also recorded over time and used to calculate volumetric water contents (θ). The paired data points were used, in conjunction with fitting software (Shypfit2.0; Peters and Durner, 2015; Pertassek et al., 2015) to construct high-resolution soil water retention curves, $\theta(h)$ (Peters and Durner, 2008; Peters et al., 2015). This software uses a modified version of the evaporation method (Schindler et al., 2010) that accounts for the non-linear water content distribution in coarse-textured samples, such as the Flint sand. Bias in the estimated retention function is caused by the uneven distribution of water at hydrostatic equilibrium in such samples. This error was avoided by using the integral evaluation of the measurements (Peters and Durner, 2008).

For fine-textured soils, such as the Hamblen silt loam, the Hyprop2 device cannot measure the entire range of soil capillary pressure heads needed to construct a complete $\theta(h)$. For such soils, the dry range of the measurement can be measured using a water activity meter (Campbell et al., 2017). Therefore, water activity meter measurements were performed on the Hamblen silt loam soil. First, ~1 g of soil was sub-sampled from the borehole material removed from the water-saturated silt loam cores. A thin layer of the moist soil was uniformly spread on the bottom in a cylindrical sample container. The soil was then exposed to the atmosphere for approximately 1-h intervals. Following each drying period, the container was capped to stop further evaporative losses. After equilibration, the container was open and placed in a water activity meter (Novasina Labmaster-a_w; Novasina-Ag, Lachen, Switzerland) to measure the activity of the soil water. After the measurement was recorded, each sample was capped and then weighed to measure soil moisture content. The process was repeated until there were no changes in the water content over time. The soil samples were then oven-dried at 105 °C for 24 h. The oven dry weights were used, along with the soil bulk density values, to determine volumetric water contents.

The water activity meter measurements were converted from water activity to total water potential (ψ_t). The ψ_t measurements were first corrected for calibration error. A best-fit line was fitted to measured calibration points determined from salt standards. The equation of the best-fit line was used to correct all of the measured ψ_t values. The water activity method includes both matric potential (ψ_m) and osmotic potential at saturation (ψ_o) when measuring ψ_t. Total potential is dominated by ψ_o in the wet range leading to erroneous measurements of ψ_m. To correct for this error, the osmotic potential at saturation was calculated and removed from the total potential based on the manufacturer's guidelines (https://www.metergroup.com/environment/articles/create-full-moisture-release-curve-using-wp4c-hyprop/) as follows:

$$\psi_m = \psi_t - \psi_o\left(\frac{\theta}{\theta_s}\right) \qquad [2]$$

The matric potential measurements were then converted to pF values. The resulting θ and pF data were added to the Hyprop2 measurements to better determine the soil water retention curve near residual water content. The van Genuchten equation (van Genuchten, 1980) was used for curve fitting the $\theta(h)$ data:

$$S_e = [1 + (\alpha h)^n]^{-m} \qquad [3]$$

where $S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r}$ is the effective saturation, with θ_s and θ_r as defined previously. The α, n, and m parameters control the overall shape of the curve. The α parameter is inversely related to the air entry pressure and controls the point at which water content starts to decline rapidly with increasing capillary pressure head. The n and m parameters are related to pore-size distribution and influence the slope and inflection point of the retention curve.

As is commonly done, the following relationship was employed to reduce the number of fitting parameters: $m = 1 - \frac{1}{n}$ (van Genuchten, 1980). As a result, Eq. [3] includes four parameters. The first parameter, the saturated water content, θ_s, was a known value for every soil core; θ_s was calculated from each sample's bulk density using an assumed specific gravity of 2.65. The other three parameters α, $n\left(=\frac{1}{1-m}\right)$, and θ_r were best estimates obtained by fitting the van Genuncten equation to the data using the Shypfit2.0 software (see Section 3.10).

2.9. Unsaturated hydraulic conductivity

The Hyprop2 device provides measurements of the water flux at the center of the soil core and the mean gradient of the hydraulic head over time. These measurements were used to construct $K(\theta)$ curves based on the Darcy-Buckingham Equation (Pertassek et al., 2015).

The unsaturated hydraulic conductivity data were fitted using the capillary conductivity function from the Mualem pore bundle model (Mualem, 1976):

$$K = \left(K_{sat}{}^{-1}\theta\right)^{\tau}\left[\left(1 - \left(K_{sat}{}^{-1}\theta\right)^{n}\right)^{\frac{1}{n}}\right]^2 \qquad [4]$$

where τ is an empirical parameter representing pore tortuosity and pore connectivity (Peters and Durner, 2015).

2.10. Data analysis

Equations [3] and [4] were fitted to the experimental $\theta(h)$ and $K(\theta)$ data sets simultaneously using the Shypfit2.0 program (Peters and Durner, 2015; Pertassek et al., 2015). Shypfit2.0 uses a non-linear regression algorithm to minimize the sum of the weighted square residuals between the model predictions and the actual data. The simultaneous fitting approach yielded a single estimate of n for both curves. Initially, the $K(\theta)$ data were fitted using the measured K_{sat} values. However, these values resulted in poor overall fits. Therefore, K_{sat} in Eq. [4] was treated as a fitting parameter, along with τ, when fitting the $K(\theta)$ data in conjunction with the fitting parameters from Eq. [3].

The root mean square error, RMSE, was calculated to determine the goodness of fit between the model predictions and observed values for both $\theta(h)$ and $K(\theta)$. The RMSE values for Flint sand ranged from 0.002 to 0.011 for the $\theta(h)$ curves, and between 0.067 and 0.372 for the $K(\theta)$ curves. For Hamblen silt loam, the RMSE values ranged from 0.003 to 0.016 for the $\theta(h)$ curves, and between 0.111 and 0.236 for the $K(\theta)$ curves. To provide an initial visual assessment of goodness of fit, the predicted $\theta(h)$ curves that yielded the median RMSE values for Flint sand (0.0038) and Hamblen silt loam (0.0095) have been plotted, along with the observed data, in Fig. 2. Similarly, the predicted $K(\theta)$ curves that yielded the median RMSE values for Flint sand (0.2872) and Hamblen silt loam (0.1479) are shown along, with the observed data, in Fig. 3. In each case, half of the 42 fits were better than those shown in Figs. 2 and 3, and half of them were worse. Note that for the Flint sand, the retention curve shows that the water content drops precipitously from saturation to ~10% water content (Fig. 2a). This occurs so quickly that the Hyprop system was unable to estimate the water flux and thus calculate the unsaturated hydraulic conductivity within this range (Fig. 3a). We thus used the saturated hydraulic conductivity to tie down the curve at the wet end, with $K(\theta)$ values rapidly approaching zero as the water content approaches residual saturation. So, the $K(\theta)$ curve for the sand is basically a step function with $K(\theta) = (K_{sat})$ above the air entry value and $K(\theta) = 0$ at residual saturation.

Analyses of variance, ANOVA, was conducted on the various fitted and measured parameters. Tukey's honestly significant difference (HSD) test was employed to identify differences between the treatment means. These analyses were performed on the measured K_{sat} values, unsaturated hydraulic conductivity curve parameters, water retention curve parameters, total plant biomass, and R_v. The α, K_{sat}, and R_v datasets were log-normally distributed and so they were $\log_{10}$-transformed prior to statistical analysis. All analyses were conducted using the SAS software package (SAS Institute Inc, 2012) with statistical significance assessed at the $p < 0.05$ level. Data from this project are freely available for download (Marcacci 2020).

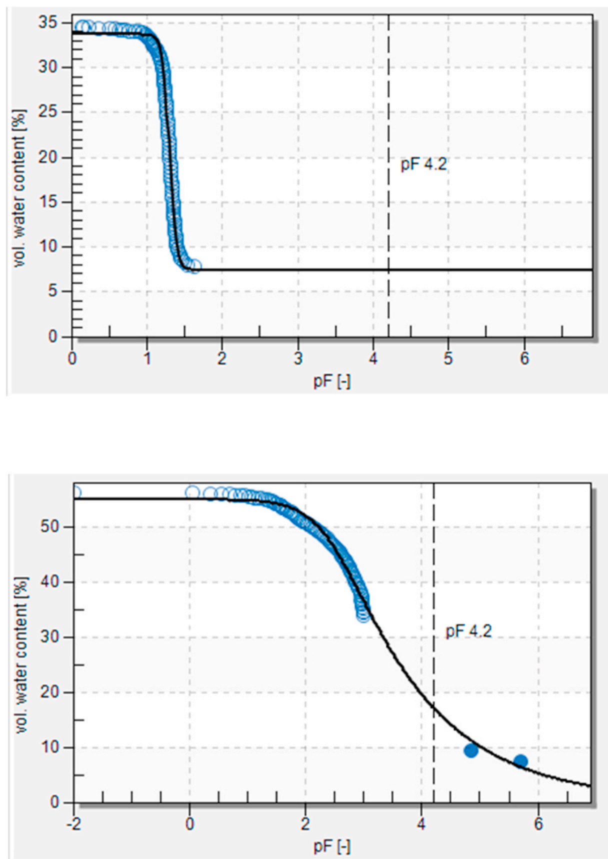

Fig. 2. Example raw data from observed (open circles) and predicted (solid lines) soil water retention curves generated from automated lab-based measurements of packed cylinders of: (A) Flint sand and (B) Hamblen silt loam. Data show the median RMSE curves out of 30 or 12 total curves produced, respectively. pF on the x-axis is the base-ten logarithm of the capillary pressure head in cm. Ref. Table S2 for fitted data to each individual replicate soil water retention curve.

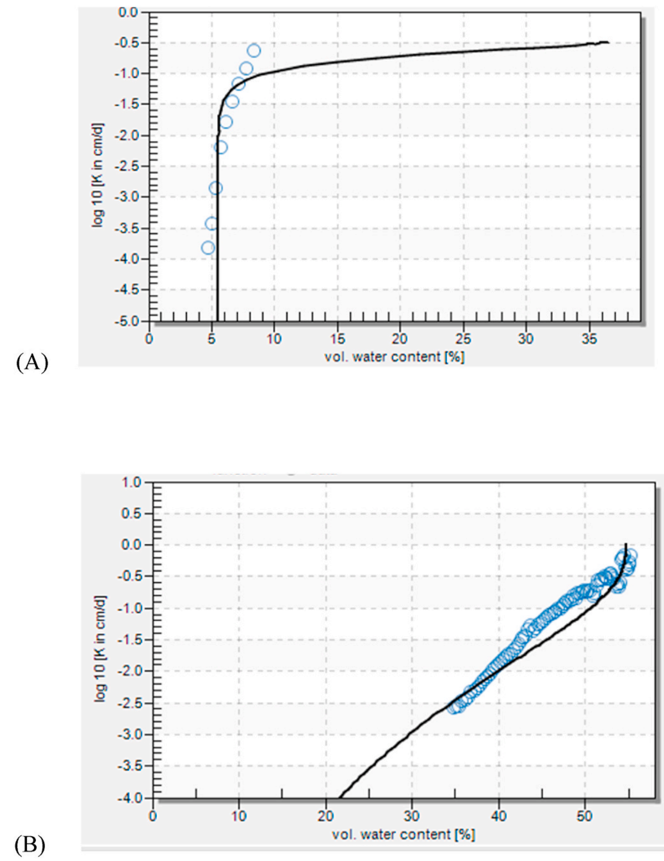

Fig. 3. Example raw data from observed (open circles) and predicted (solid lines) unsaturated soil hydraulic conductivity curves generated from automated lab-based measurements of packed cylinders of: (A) Flint sand and (B) Hamblen silt loam. Data show the median RMSE curves out of 30 or 12 total curves produced, respectively.

3. Results

3.1. Switchgrass biomass

Total biomass of the switchgrass plants grown in the Flint sand was analyzed using a two-way ANOVA. There were significant treatment and growth cycle effects, but no significant interaction between these terms. Thus, the results were combined so that treatment effects were averaged across the two growth cycles. A Tukey's HSD test showed that the total harvested shoot and root biomass in the FRT and RTS + MYC treatments was significantly greater than in the RTS treatment. Yet when analyzed separately, there was no difference in shoot biomass among the treatments ($p = 0.066$), and RTS + MYC had significantly greater root biomass than RTS or FRT (Fig. 4 A, B).

For the Hamblen silt loam, the mean total shoot and root biomass for the RTS + MYC treatment was over ten times greater than for the RTS treatment (Fig. 4 C, D). However, there was substantial variation among replicate cores within the RTS + MYC treatment. As a result, a one-sided *t*-test performed on these data indicated no significant difference ($p =$ 0.07–0.08) between the RTS and RTS + MYC treatments for the Hamblen silt loam.

3.2. Switchgrass root volume ratio

A two-way ANOVA was run on the $\log_{10} R_v$ values for plants grown in the Flint sand. Results showed a significant treatment effect, but no significant growth cycle or treatment × growth cycle effects. Thus, the data from both growth cycles were combined. Post-hoc analysis indicated that R_v for the RTS + MYC treatment was significantly higher than that for the RTS treatment, while the FRT treatment was not significantly different from either of those treatments (Fig. 5A). A similar trend was observed for the $\log_{10} R_v$ values for plants grown in the Hamblen silt loam where R_v for the RTS + MYC treatment was statistically greater than for the RTS treatment (Fig. 5B).

3.3. Saturated hydraulic conductivity

The measured $\log_{10} K_{sat}$ values for Flint sand revealed significant effects of treatment and growth cycle, with no significant interaction. The RTS treatment was significantly lower than that of the CON treatment (Fig. 6). The other treatments (MYC, FRT, RTS + MYC), were not significantly different from either of these two treatments. The measured K_{sat} values for the Hamblen silt loam were between 2 and 4 orders of magnitude lower than those for the Flint sand, and there was no difference in $\log_{10} K_{sat}$ between any of the treatments at the 95% confidence interval (results not shown). Measured K_{sat} values for each individual core are displayed in Table S1.

3.4. Water retention curve parameters

Two-way and one-way ANOVA's were run on the soil water retention curve parameters for Flint sand and Hamblen silt loam, respectively. The treatment × growth cycle interaction was not significant in any of the two-way ANOVA's and so the treatment effect for Flint sand was

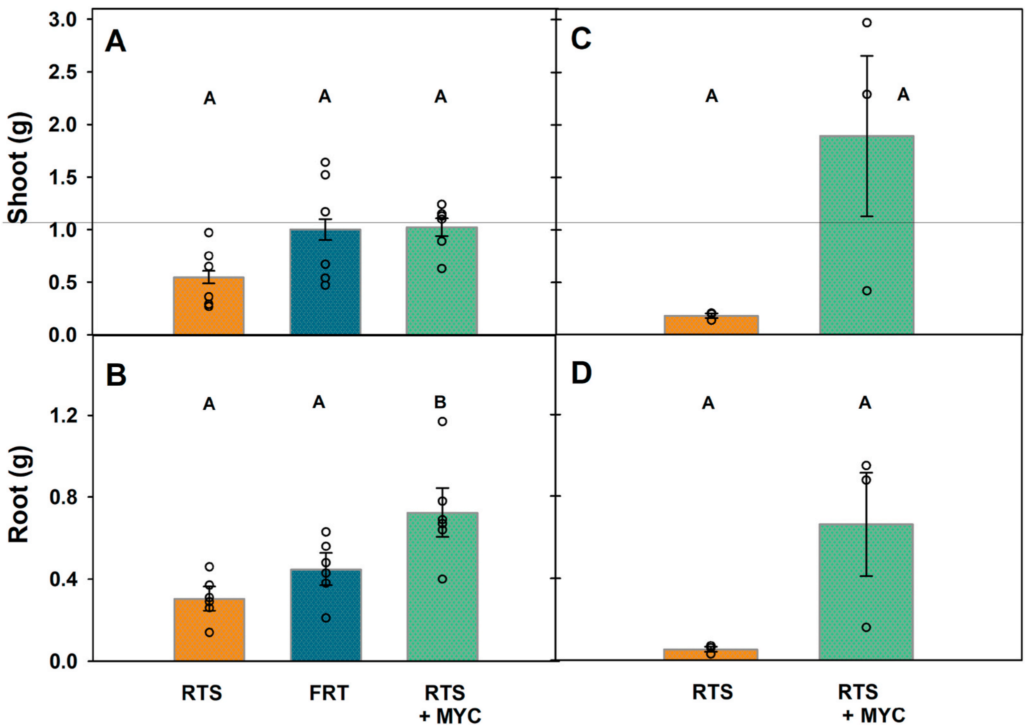

Fig. 4. Switchgrass total shoot and root dry biomass for rooted (switchgrass only; RTS), fertilized rooted (fertilized switchgrass; FRT) or mycorrhizal rooted (switchgrass inoculated with an endophytic fungus, *S. indica*; RTS + MYC) chambers when grown in a greenhouse in (A, B) Flint sand for 70 days or (C, D) Hamblen silt loam for 80 days. Data show individual data points and mean ± SE. Treatments that share a letter within a panel are not significantly different at $p < 0.05$ based on Tukey HSD.

evaluated by averaging over both growth periods. The results of these ANOVA's are summarized in Table 3. It should be noted that the θ_r parameter for Hamblen silt loam soil is not included in Table 3. This is because model estimates of θ_r were zero for this soil, likely due to the inclusion of the water activity measurements at the dry end of the curves (e.g., Fig. 2B).

The only significant treatment effects observed were with θ_s and n for the Flint sand (Table 3). For θ_s, the CON treatment had the lowest mean value, sequentially followed by the CON + MYC, RTS, FRT, and RTS + MYC treatments (Fig. 7). The opposite tendency was observed with the pore-size distribution fitting parameter n (Fig. 8). The mean values for this parameter followed a trend whereby the CON treatment had the highest mean value, and the RTS + MYC treatment had the lowest mean value. An almost identical ranking across the treatments was obtained in the mean n values for the Hamblen silt loam; however, differences were not significant ($p = 0.062$; Table 3; Fig. 8B).

3.5. Unsaturated hydraulic conductivity curve parameters

The tortuosity τ parameter and log-transformed K_{sat} values estimated by fitting Eq. [4] to the $K(\theta)$ data were statistically analyzed using two-way and one-way ANOVA's for Flint sand and Hamblen silt loam, respectively. Regardless of soil type there were no significant treatment effects for either parameter. No clear trends were visible in the means of the estimated $\log_{10} K_{sat}$ values, which is in stark contrast to the measured $\log_{10} K_{sat}$ values (see Fig. 6), and emphasizes the fact that saturated hydraulic conductivity was simply a fitting parameter in the context of the $K(\theta)$ results. Although not statistically significant, the mean τ values for the two mycorrhizal treatments were consistently higher than the mean values for the other treatments for both soils (results not shown). Soil hydraulic parameter values for each individual core are displayed in Table S2.

4. Discussion

4.1. Switchgrass biomass

For Flint sand, the non-mycorrhizal rooted switchgrass plants had less root biomass, total biomass and root volume than plants inoculated with the endophyte, *S. indica*. The presence of mycorrhizal fungi thus increased total plant growth in the sand, equivalent to the addition of fertilizer, in comparison to the uninoculated plants. As mutualistic symbionts, mycorrhizal fungi have been shown to increase access to nutrients that the plant roots could not acquire otherwise (Hetrick, 1991), even in higher fertility soils (Van Der Heijden et al., 2006). While not significant, the biomass results for Hamblen silt loam mirrored those for Flint sand, with the mean values for roots and shoots of the rooted treatment much less than the mean value for the mycorrhizal rooted treatment. In both Flint sand and Hamblen silt loam, the presence of mycorrhizal fungi enabled plants to grow more biomass than the non-mycorrhizal treated samples, which led to larger plants, and should lead to a larger effect on soil hydraulic properties. The effect of mycorrhizal fungi such as *S. indica* on total plant biomass growth has implications for enhancing sustainable agriculture practices, particularly on nutrient poor soils, where inoculations might help decrease reliance on fertilizer applications (Berruti et al., 2016; Dias et al., 2020).

The presence of mycorrhizal fungi increased root growth, thereby significantly increasing the root volume ratio in both soil types. As R_v is a primary component in calculating void ratio in Eq. [1], the results imply that the mycorrhizal roots filled more total pore space than the untreated roots. The ability of mycorrhizal fungi to access nutrients may have caused the inoculated roots to focus growth on primary elongation and radial expansion (Hetrick, 1991; Kuyper et al., 2021). This led to visually longer (in the vertical direction) and radially-thicker roots associated with the mycorrhizae-inoculated plants. The longer and thicker roots rearranged soil particles and aggregates as the roots radially expanded into available macropores. In contrast, the non-mycorrhizal nutrient-limited plants tended to grow thinner, finer roots with more lateral offshoots and root hairs, as supported by earlier root structural work (López-Bucio et al., 2003). In a study of 12 cover

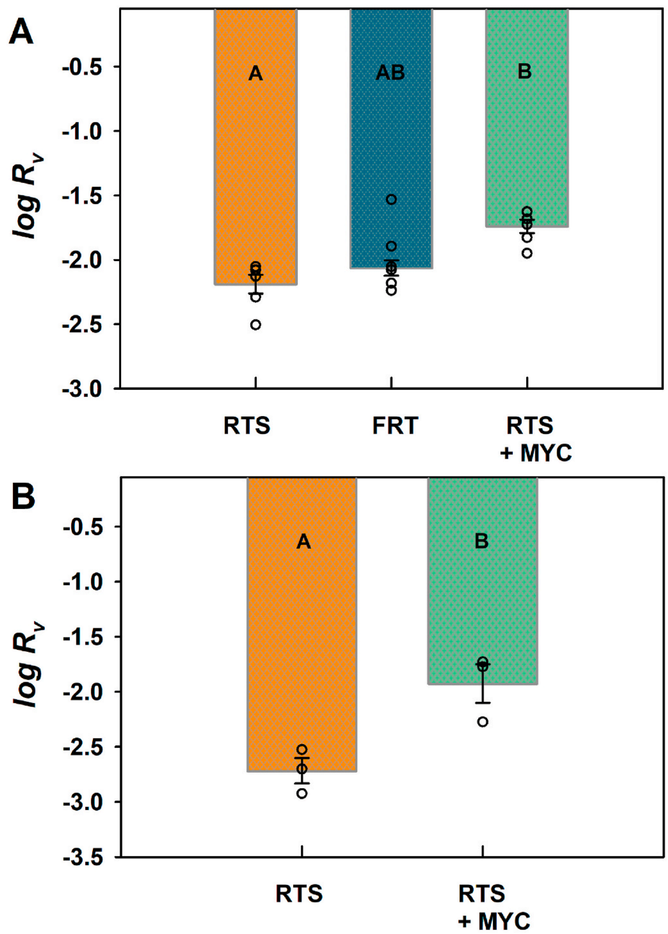

Fig. 5. Log-transformed root volume ratio (log R_v) for switchgrass grown in packed cylinders of (A) Flint sand for 70 days and (B) Hamblen silt loam for 80 days with either switchgrass (RTS), fertilized switchgrass (FRT) or switchgrass inoculated with endophytic fungi (RTS + MYC). Data show mean ± SE. Treatments that share a letter within a panel are not significantly different at $p < 0.05$ based on Tukey HSD.

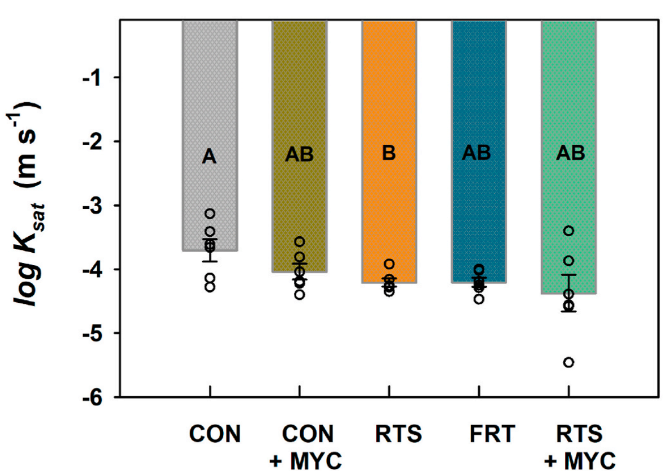

Fig. 6. Log-transformed saturated hydraulic conductivity (log K_{sat}) for Flint sand with or without switchgrass and mycorrhizae; sand only (CON), *S. indica* in sand (CON + MYC), switchgrass in sand (RTS), fertilized switchgrass in sand (FRT), switchgrass and *S. indica* in sand (RTS + MYC). Data show mean ± SE. Treatments that share a letter are not significantly different at $p < 0.05$ based on Tukey HSD.

Table 3
Summary of ANOVA results for the soil water retention curve fitting parameters from Equation [3] for Flint sand (FS) and Hamblen silt loam (HSL).

Soil Type	Parameter	Model R^2	Model F-value	Model p-value	Treatment F-Value	Treatment p-value
FS	θ_s	0.8356	24.39	<.0001	27.17	<.0001
FS	$\log_{10} \alpha$	0.1523	1.120	0.3681	1.120	0.3681
FS	n	0.8164	27.79	<.0001	27.79	<.0001
FS	θ_r	0.1241	0.890	0.4869	0.890	0.4869
HSL	θ_s	0.0133	0.430	0.7365	0.430	0.7365
HSL	$\log_{10} \alpha$	0.4225	1.950	0.2001	1.950	0.2001
HSL	n	0.5868	3.790	0.0586	3.790	0.0586

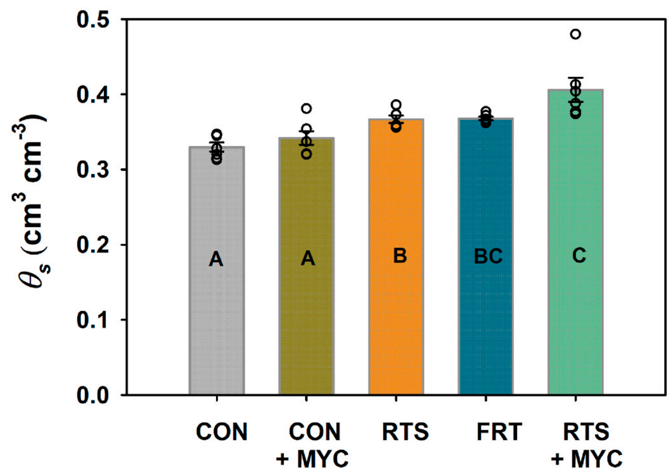

Fig. 7. Saturated volumetric water content (θ_s) for Flint sand with or without switchgrass and mycorrhizae; sand only (CON), *S. indica* in sand (CON + MYC), switchgrass in sand (RTS), fertilized switchgrass in sand (FRT), switchgrass and *S. indica* in sand (RTS + MYC). Data show mean ± SE. Treatments that share a letter are not significantly different at $p < 0.05$ based on Tukey HSD.

crops species that varied in root traits, root diameter was shown to have strong relationships with pore size, highlighting the ability to manage for desired pore distributions in the field (Bodner et al., 2014). Their work provides a framework for scaling root trait induced changes at the pore scale to macro-scale prediction of hydraulic properties that can be applied to the landscape. This highlights the need for additional studies that link specific root traits and distribution (e.g., Rajamanthri et al., 2021) to soil hydraulic properties that can be scaled to the landscape level based on the specific characteristics of the vegetation.

4.2. Saturated hydraulic conductivity

The constant head flow experiments conducted on Flint sand in this study verified previous research (Barley, 1954; Leung et al., 2015a; Scanlan and Hinz, 2010), which showed the presence of living roots decreased K_{sat} in comparison to unvegetated soil. It was found that the K_{sat} measurements for the Flint sand soil supported the premise of our first hypothesis, *i.e.*, the presence of roots resulted in statistically lower mean K_{sat} values in comparison to the control (Fig. 6). In a highly controlled study conducted earlier, K_{sat} of root and hyphal-free Flint sand was found to be ~1.7×10^{-2} cm s^{-1} (Kang et al., 2014), overlapping with results from control chambers in this study. However, when that value was used to model soil hydraulics in the presence of roots, it was much too high to fit the observed data (Dhiman et al., 2018), which thus further supports the decline in K_{sat} with roots and mycorrhizae as described in this study.

Although K_{sat} was not statistically different between treatments for the natural silt loam soil, the results do offer insight into the intricate

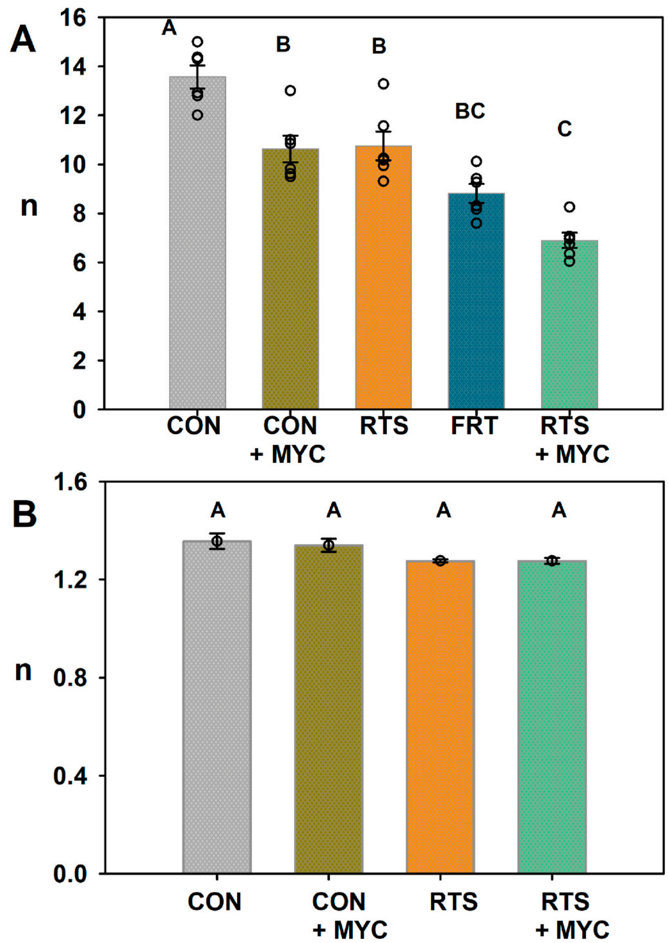

Fig. 8. Pore-size distribution fitting parameter (n) for (A) Flint sand and (B) Hamblen silt loam with or without switchgrass and mycorrhizae; sand only (CON), *S. indica* in sand (CON + MYC), switchgrass in sand (RTS), fertilized switchgrass in sand (FRT), switchgrass and *S. indica* in sand (RTS + MYC). Data show mean ± SE. Treatments that share a letter are not significantly different at $p < 0.05$ based on Tukey HSD.

relationships between soil, soil biota and K_{sat}. For this soil, measured K_{sat} from control cores were tightly grouped, but differential success of the fungi and roots in the mycorrhizal and mycorrhizal root-permeated treatments resulted in K_{sat} variation by 1–2 orders of magnitude across replicates. While the variation indicates the importance of roots and mycorrhizal fungi on saturated hydraulic conductivity, it also limited detection of significant differences across treatments. The differences in magnitude and disparity in the K_{sat} results between the two soils were also likely due to the differences in textures and nutrient status. The repacked Hamblen soil was fine-textured and had few macropores, whereas the coarse-grained Flint sand had many macropores. As a result, roots and fungal hyphae were able to readily fill the macropores in Flint sand and reduce its saturated hydraulic conductivity.

As a root develops, the tip of the growing root is forced into similarly-sized soil pores. Over time, the growing root swells and causes a shift in the arrangement of soil particles (Hillel, 2004). This process either increases soil porosity (φ) and thereby reduces bulk density or causes soil pores to become plugged, thereby reducing φ. The lower K_{sat} values for the rooted treatment in the Flint sand soil suggested that the roots reduced macroporosity via pore plugging (Sedgley and Barley, 1954). Similar results were reported for another grass species in both a sandy clay and a silt soil, likely by roots clogging pores and changing soil structure (Jotisankasa and Sirirattanachat, 2017).

The growth pattern of roots is also dependent on nutrient availability, which varied across our treatments. The greater lateral growth observed with the rooted control treatment likely also increased pore blocking in comparison to the FRT and RTS + MYC treatments. The pore-blocking effect was not seen in the fertilizer treatment due to the addition of supplementary nutrients. Additionally, the pore blocking effect was not seen in the mycorrhizal rooted treatment likely due to the ability of smaller diameter hyphae to access additional nutrients not directly available to the much larger roots; e.g., those nutrients within micropores. In the Hamblen silt loam, the lack of treatment differences was likely due to the increased availability of nutrients in the fine-textured soil, that likely reduced demand for carbon allocation to root growth, and was reflected in the subsequent reduction in root volume compared with the Flint sand. With access to available nutrients, the rooted treatment from the silt loam soil did not have to sacrifice primary root elongation for lateral root growth. The lack of substantial pore plugging resulted in little difference in mean K_{sat} values between the planted and unplanted treatments in this soil.

4.3. Soil water retention curve

The shape of the soil water retention curve, $\theta(h)$, is dependent upon the relationship between soil porosity and vegetation (Ni et al., 2018). Our second hypothesis stated that the presence of roots and mycorrhizal fungi would change the shape of the $\theta(h)$ near saturation. Indeed, the RTS, FRT, and RTS + MYC treatments all increased θ_s in Flint sand soil relative to the controls with no roots, in agreement with earlier studies of vegetated soil (e.g., Bodner et al., 2014; Rajamanthri et al., 2021). Assuming complete water saturation (i.e., no trapped air), the θ_s is equivalent to the porosity. Therefore, roots increased the φ as they grew in the Flint sand. In consolidated porous media, growing roots would clog the open pore space and cause a reduction in θ_s and φ. In the unconsolidated Flint sand, however, the growing roots caused a shift in the arrangement of grains within the rhizosphere. The growing roots thus caused an increase in φ as the soil volume expanded. Equation [1] predicts that the presence of roots will reduce the soil porosity. However, our θ_s results for Flint sand suggest the opposite was true in these experiments (Fig. 7). Even so, the patterns of root distribution, growth and turnover are likely to lead to dynamic changes in φ over time for loose, sandy soils.

The α parameter is inversely related to the air entry point of the soil. This parameter controls the shape of the $\theta(h)$ by defining the point at which water content starts to decline rapidly with increasing pressure head. There was no significant differences in the α parameter between treatments for either soil, which is a bit surprising as α is often interpreted as a proxy parameter for macroporosity. Prior studies have shown mixed results (Leung et al., 2015b; Shao et al., 2017) indicating root impacts on this parameter may be dependent on study conditions, such as rooting density.

The n parameter characterizes the pore-size distribution. Higher n values denote steeper slopes of the $\theta(h)$ curve, and narrower pore-size distributions, while lower n values indicate more gradual slopes and broader pore-size distributions (Bodner et al., 2014; Rachman et al., 2004). For the Flint sand, the unrooted controls had the highest n values and, consequently, the steepest slopes. Slope steepness decreased in the CON + MYC, RTS, FRT and RTS + MYC treatments, indicating that roots and fungal hyphae expanded the pore-size distribution, similar to results from another endophytic fungal inoculated soil (Hosseini et al., 2016). Although not statistically significant for the Hamblen silt loam, the mean n values for this soil did share a similar trend to the mean values for the Flint sand treatments.

The trends discussed above for the Flint sand and, to a lesser extent, the Hamblen silt loam, support the second hypothesis. The presence of roots and roots with mycorrhizal fungi increased φ and decreased the n parameter and thereby changed the shape of the soil water retention curve when compared to bare soil. In the case of both θ_s and n, the mycorrhizal rooted treatment was significantly different from the rooted treatment alone, thereby supporting our fourth hypothesis. Mycorrhizal fungi caused a broadening of the pore-size distribution through the promotion of enhanced root growth, as outlined above. Interestingly, the present study appears to be relatively novel in finding a strong effect of roots and mycorrhizae on the porosity and slope of the $\theta(h)$ curve. Most previous studies have reported strong statistical differences amongst α values, with roots and mycorrhizae increasing or decreasing α, but we did not observe this effect. For the high-porosity Flint sand, this may be due to incomplete root filling of pore space such that the air entry point was minimally affected; for the Hamblen silt loam, this was likely due to the high variation in fitting results for α for the roots + mycorrhizae samples.

The residual water content (θ_r) is the amount of water remaining in the soil at very high capillary pressure heads. For Flint sand, the θ_r parameter showed no significant differences between treatments. Information on θ_r is rarely reported in the literature; in one of the few studies to investigate the effects of roots on residual water content, Leung et al. (2015b) found that θ_r was consistent across root-permeated and bare soil, supporting our observation of lack of sensitivity for this parameter.

4.4. Unsaturated hydraulic conductivity

The present study appears to be one of the first to compare K(θ) parameters between bare, rooted, and mycorrhizal rooted soils. However, evidence showed that neither roots, nor roots with mycorrhizal fungi, had any pronounced influence on unsaturated flow parameters in either Flint sand or Hamblen silt loam under the conditions of these experiments. As a result, there was no support for our third hypothesis, which stated that root and mycorrhizal hyphae derived changes in pore connectivity would increase the unsaturated hydraulic conductivity of the soil. The presence of roots was expected to decrease the amount of open preferential flow paths via pore-clogging and therefore skew open pore spaces toward smaller pore radii. The lack of statistical significance in the silt loam was perhaps predictable; as a fine-textured soil, the presence of roots and mycorrhizal fungi generally show less impact on soil structure overall (Lehmann et al., 2017). The lack of statistical significance also likely reflects the high degree of variability in $K(\theta)$ found in nature.

5. Concluding remarks

This research investigated the influence of roots and mycorrhizal fungi on the flow and retention of water in soil through the statistical comparison of measured and fitted hydraulic parameters. The influence of vegetation on soil hydraulic parameters was explored through the application of laboratory experiments and quantification of switchgrass biomass and root volume ratios (R_v). Using this approach, statistically significant relationships were found between the biological measurements and soil hydraulic properties.

In Flint sand, roots with mycorrhizal fungi had a greater influence on hydraulic properties than roots alone. The presence of roots with mycorrhizal fungi increased root biomass, which was reflected in increased R_v values. Root volume was similarly increased by mycorrhizal fungi in the Hamblen silt loam. Roots increased porosity in the sand, while simultaneously clogging pore spaces and preventing preferential water flow. This effect of plant roots on porosity was reflected in the shape of the soil water retention curve, where the presence of roots increased θ_s and decreased n, and mycorrhizal fungi further accentuated these effects. The presence of roots with mycorrhizal fungi changed the shape of the $\theta(h)$ for Flint sand in comparison to roots, with mycorrhizal fungi moving the curve upward by increasing ϕ and changing the pore-size distribution, thereby decreasing the slope. The influence of roots and mycorrhizal fungi on soil hydraulic properties was less obvious in the Hamblen silt loam, as none of the treatments were statistically different from each other.

This research has several limitations worth noting. Due to the nature of the relatively short growth periods, the effect of roots and mycorrhizal fungi on soil hydraulic properties was limited by time. Literature has shown that the effect of roots on pore plugging is time-dependent with the aging and death of roots opening root-created macropores and increasing saturated flow via preferential flow paths (Leung et al., 2015b; Ng et al., 2016; Scholl et al., 2014). Additionally, the research is limited to assessment of just two contrasting soil types, with one plant species. Extension of this research to a wider range of plants and soil types would be valuable.

This study also focused on the impacts of a specific mycorrhizal fungus and its association with switchgrass on soil hydraulic properties; however, there is potential for different mycorrhizal-plant associates to have alternative results. Multiple sub-species of mycorrhizal fungi can associate with a plant simultaneously, including endomycorrhizas and ectomycorrhizas that vary widely in their functional and structural traits. Separate species have been known to compete for access to resources and mutualistic plant assimilates. Therefore, a multi-mycorrhizal association could affect root-hyphal impacts on soil hydraulic properties differently, spatially within the soil profile, and would be worth investigating. Further research into the hydraulic properties of mycorrhizal-plant associates grown in disturbed and undisturbed natural soil conditions is also warranted.

As models often use pedotransfer functions to estimate soil hydraulic properties from soil texture, and sometimes bulk density and organic matter content, they might be further refined by inclusion of root-mycorrhizal impacts on those properties. Current models that include root-influences on soil hydraulic properties usually leverage only root volume ratio (e.g., Assouline and Or, 2014; Ng et al., 2016), although root size distribution (Bodner et al., 2014) and mycorrhizal association should also be considered. As results continue to accumulate, their application to soil hydrology components of earth system models can be used to refine sensitivity of vegetation productivity to hydraulic properties and soil water availability at larger scales.

Notice: This manuscript has been authored by UT-Battelle, LLC under Contract No. DE-AC05-00OR22725 with the US Department of Energy. The United States Government retains and the publisher, by accepting the article for publication, acknowledges that the United States Government retains a non-exclusive, paid-up, irrevocable, world-wide license to publish or reproduce the published form of this manuscript, or allow others to do so, for United States Government purposes. The Department of Energy will provide public access to these results of federally sponsored research in accordance with the DOE Public Access Plan (http://energy.gov/downloads/doe-public-access-plan).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Instrumentation for measuring soil hydraulic properties was generously provided by the METER Group Inc. through a Grant A. Harris Fellowship awarded to K.M. Marcacci. E. Perfect acknowledges support from the Tom Cronin and Helen Sestak Faculty Achievement award. We appreciate laboratory assistance from Claire Bottini, Joanne Childs and Ariel Ebanks. This material is based upon work supported by the U.S. Department of Energy, Office of Science, Office of Biological and Environmental Research. ORNL is managed by UT-Battelle, LLC, for the DOE under contract DE-AC05-00OR22725.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.

org/10.1016/j.rhisph.2022.100510.

References

Adams, S.B., Kszos, L.A., 2005. Development of switchgrass (*Panicum virgatum*) as a bioenergy feedstock in the United States. Biomass Bioenergy 28 (6), 515–535. https://doi.org/10.1016/j.biombioe.2004.05.006.

Ahmed, M.A., Kroener, E., Benard, P., Zarebanadkouki, M., Carminati, A., 2016. Drying of mucilage causes water repellency in the rhizosphere of maize: measurements and modelling. Plant Soil 407, 161–171. https://doi.org/10.1007/s11104-015-2749-1.

Akhzari, D., 2015. Response of *Glycyrrhiza glabra* L. to arbuscular mycorrhizal fungi and water stress. J. Essent. Oil Bear. Plant 18, 992–1002. https://doi.org/10.1080/0972060x.2014.890085.

Aravena, J.N.E., Berli, M., Ghezzehei, T.A., Tyler, S.W., 2011. Effects of root-induced compaction on rhizosphere hydraulic properties - X-ray microtomography imaging and numerical simulations. Environ. Sci. Technol. 45, 425–431. https://doi.org/10.1021/es102566j.

Assouline, S., Or, D., 2014. The concept of field capacity revisited: defining intrinsic static and dynamic criteria for soil internal drainage dynamics. Water Resour. Res. 50, 4787–4802. https://doi.org/10.1002/2014wr015475.

Augé, R.M., Stodola, A.J.W., Tims, J.E., Saxton, A.M., 2001. Moisture retention properties of a mycorrhizal soil. Plant Soil 230, 87–97. https://doi.org/10.1023/A:1004891210871.

Augé, R.M., Sylvia, D.M., Park, S., Buttery, B.R., Saxton, A.M., Moore, J.L., Cho, K., 2004. Partitioning mycorrhizal influence on water relations of *Phaseolus vulgaris* into soil and plant components. Can. J. Bot. 82, 503–514. https://doi.org/10.1139/b04-020.

Barley, K.P., 1954. Effects of root growth and decay on the permeability of a synthetic sandy loam. Soil Sci. 78, 205–210. https://doi.org/10.1097/00010694-195409000-00005.

Berruti, A., Lumini, E., Balestrini, R., Bianciotto, V., 2016. Arbuscular mycorrhizal fungi as natural biofertilizers: let's benefit from past successes. Front. Microbiol. 6, 1559. https://doi.org/10.3389/fmicb.2015.01559.

Bitterlich, M., Franken, P., Graefe, J., 2018. Arbuscular mycorrhiza improves substrate hydraulic conductivity in the plant available moisture range under root growth exclusion. Front. Plant Sci. 9 https://doi.org/10.3389/fpls.2018.00301.

Bodner, G., Leitner, D., Kaul, H.P., 2014. Coarse and fine root plants affect pore size distributions differently. Plant Soil 380, 133–151. https://doi.org/10.1007/s11104-014-2079-8.

Brito, I., de Carvalho, M., Goss, M.J., 2009. Techniques for arbuscular mycorrhiza inoculum reduction. In: Symbiotic Fungi, vol. 18. Soil Biology, pp. 307–318. https://doi.org/10.1007/978-3-540-95894-9_19.

Campbell, G.S., Campbell, C.S., Rivera, L., D R, C., B, C.L., Chambers, C., 2017. How to Generate a Soil Moisture Characteristic Using the WP4C. Meter Environment Knowledge Base Meter Group Munich, Germany.

Carminati, A., Moradi, A.B., Vetterlein, D., Vontobel, P., Lehmann, E., Weller, U., Vogel, H.-J., Oswald, S.E., 2010. Dynamics of soil water content in the rhizosphere. Plant Soil 332, 163–176. https://doi.org/10.1007/s11104-010-0283-8.

Carminati, A., 2012. A model of root water uptake coupled with rhizosphere dynamics. Vadose Zone J. 11 https://doi.org/10.2136/vzj2011.0106 vzj2011.0106.

Czarnes, S., Hallett, P.D., Bengough, A.G., Young, I.M., 2000. Root- and microbial-derived mucilages affect soil structure and water transport. Eur. J. Soil Sci. 51, 435–443. https://doi.org/10.1046/j.1365-2389.2000.00327.x.

Daly, K.R., Mooney, S.J., Bennett, M.J., Crout, N.M.J., Roose, T., Tracy, S.R., 2015. Assessing the influence of the rhizosphere on soil hydraulic properties using X-ray computed tomography and numerical modelling. J. Exp. Bot. 66, 2305–2314. https://doi.org/10.1093/jxb/eru509.

Dane, J.H., Topp, G.C., 2002. Methods of Soil Analysis: Part 4. Physical Method Soil Science Society of America, Madison, WI.

Daynes, C.N., Field, D.J., Saleeba, J.A., Cole, M.A., McGee, P.A., 2013. Development and stabilisation of soil structure via interactions between organic matter, arbuscular mycorrhizal fungi and plant roots. Soil Biol. Biochem. 57, 683–694. https://doi.org/10.1016/j.soilbio.2012.09.020.

Dexter, A.R., 1987. Compression of soil around roots. Plant Soil 97, 401–406. https://doi.org/10.1007/bf02383230.

Dhiman, I., Bilheux, H., DeCarlo, K., Painter, S.L., Santodonato, L., Warren, J.M., 2018. Quantifying root water extraction after drought recovery using sub-mm in situ empirical data. Plant Soil 424, 73–89. https://doi.org/10.1007/s11104-017-3408-5.

Dias, T., Pimentel, V., Cogo, A.J.D., Costa, R., Bertolazi, A.A., Miranda, C., de Souza, S.B., Melo, J., Carolino, M., Varma, A., Eutrópio, F., Olivares, F.L., Ramos, A.C., Cruz, C., 2020. The free-living stage growth conditions of the endophytic fungus *Serendipita indica* may regulate its potential as plant growth promoting microbe. Front. Microbiol. 11, 562238. https://doi.org/10.3389/fmicb.2020.562238.

Durner, W., 1994. Hydraulic conductivity estimation for soils with heterogeneous pore structure. Water Resour. Res. 30, 211–223. https://doi.org/10.1029/93wr02676.

Fatichi, S., Or, D., Walko, R., et al., 2020. Soil structure is an important omission in Earth System Models. Nat. Commun. 11, 522. https://doi.org/10.1038/s41467-020-14411-z.

Gallipoli, D., Wheeler, S.J., Karstunen, M., 2003. Modelling the variation of degree of saturation in a deformable unsaturated soil. Geotechnique 53, 105–112. https://doi.org/10.1680/geot.2003.53.1.105.

Gehring, C.A., 2017. Introduction: mycorrhizas and soil structure, moisture, and salinity. In: Johnson, N., et al. (Eds.), Mycorrhizal Mediation of Soil. Elsevier Inc., pp. 235–240

Guha, A., Han, J., Cummings, C.R., McLennan, D.A., Warren, J.M., 2018. Differential ecophysiological responses and resilience to heat wave events in four co-occurring temperate tree species. Environ. Res. Lett. 13 https://doi.org/10.1088/1748-9326/aabcd8.

Hallett, P., Feeney, D., Bengough, A., Rillig, M., Scrimgeour, C., Young, I., 2009. Disentangling the impact of AM fungi versus roots on soil structure and water transport. Plant Soil 314, 183–196. https://doi.org/10.1007/s11104-008-9717-y.

Hetrick, B.A.D., 1991. Mycorrhizas and root architecture. Experientia 47, 355–362. https://doi.org/10.1007/bf01972077.

Hillel, D., 2004. Introduction to Environmental Soil Physics. Elsevier Academic Press, Amsterdam.

Hosseini, F., Mosaddeghi, M.R., Hajabbasi, M.A., Sabzalian, M.R., 2016. Role of fungal endophyte of tall fescue (*Epichloë coenophiala*) on water availability, wilting point and integral energy in texturally-different soils. Agric. Water Manag. 163, 197–211. https://doi.org/10.1016/j.agwat.2015.09.024.

Jarvis, S., Woodward, S., Alexander, I.J., Taylor, A.F., 2013. Regional scale gradients of climate and nitrogen deposition drive variation in ectomycorrhizal fungal communities associated with native Scots pine. Global Change Biol. 19, 1688–1696. https://doi.org/10.1111/gcb.12178.

Jotisankasa, A., Sirirattanachat, T., 2017. Effects of grass roots on soil-water retention curve and permeability function. Can. Geotech. J. 54, 1612–1622. https://doi.org/10.1139/cgj-2016-0281.

Kang, M., Bilheux, H.Z., Voisin, S., Cheng, C.L., Perfect, E., Horita, J., Warren, J.M., 2013. Water calibration measurements for neutron radiography: application to water content quantification in porous media. Nucl. Instrum. Methods Phys. Res. 708, 24–31. https://doi.org/10.1016/j.nima.2012.12.112.

Kang, M., Perfect, E., Cheng, C.L., Bilheux, H.Z., Lee, J., Horita, J., Warren, J.M., 2014. Multiple pixel-scale water retention curves quantified by neutron radiography. Adv. Water Resour. 65 (18) https://doi.org/10.1016/j.advwatres.2013.12.004.

Kosugi, K.I., 1996. Lognormal distribution model for unsaturated soil hydraulic properties. Water Resour. Res. 32, 2697–2703. https://doi.org/10.1029/96wr01776.

Kuyper, T.W., Wang, X., Muchane, M.N., 2021. The interplay between roots and arbuscular mycorrhizal fungi influencing water and nutrient acquisition and use efficiency. In: Rengel, Z., Djalovic, I. (Eds.), The Root Systems in Sustainable Agricultural Intensification. https://doi.org/10.1002/9781119525417.ch7.

Lehmann, A., Leifheit, E.F., Rillig, M.C., 2017. Mycorrhizas and soil aggregation. In: Johnson, N.C., et al. (Eds.), Mycorrhizal Mediation of Soil Fertility, Structure, and Carbon Storage. Elsevier, p. 526.

Leung, A.K., Garg, A., Coo, J.L., Ng, C.W.W., Hau, B.C.H., 2015a. Effects of the roots of *Cynodon dactylon* and *Schefflera heptaphylla* on water infiltration rate and soil hydraulic conductivity. Hydrol. Process. 29, 3342–3354. https://doi.org/10.1002/hyp.10452.

Leung, A.K., Garg, A., Ng, C.W.W., 2015b. Effects of plant roots on soil-water retention and induced suction in vegetated soil. Eng. Geol. 193, 183–197. https://doi.org/10.1016/j.enggeo.2015.04.017.

López-Bucio, J., Cruz-RamíRez, A., Herrera-Estrella, L., 2003. The role of nutrient availability in regulating root architecture. Curr. Opin. Plant Biol. 6, 280–287. https://doi.org/10.1016/s1369-5266(03)00035-9.

Marcacci, K.M., 2020. Influence of Living Plant Roots and Mycorrhizal Hyphae on Soil Hydraulic Properties. Master's Thesis, University of Tennessee. https://trace.tennessee.edu/utk_gradthes/6261, 2020.

Moll, J., Hoppe, B., König, S., Wubet, T., Buscot, F., Krüger, D., 2016. Spatial distribution of fungal communities in an arable soil. PLoS One 11 (2), e0148130. https://doi.org/10.1371/journal.pone.0148130.

Mualem, Y., 1976. A new model for predicting the hydraulic conductivity of unsaturated porous media. Water Resour. Res. 12, 513–522. https://doi.org/10.1029/wr012i003p00513.

Müller, A., Ngwene, B., Peiter, E., et al., 2017. Quantity and distribution of arbuscular mycorrhizal fungal storage organs within dead roots. Mycorrhiza 27, 201–210. https://doi.org/10.1007/s00572-016-0741-0.

Ng, C.W.W., Ni, J.J., Leung, A.K., Wang, Z.J., 2016. A new and simple water retention model for root-permeated soils. Géotech. Lett. 6, 106–111. https://doi.org/10.1680/jgele.15.00187.

Ni, J.J., Leung, A.K., Ng, C.W.W., 2018. Modelling effects of root growth and decay on soil water retention and permeability. Can. Geotech. J. 56, 1049–1055. https://doi.org/10.1139/cgj-2018-0402.

NRCS, 2019. Soil Survey of Roane County. United States Department of Agriculture, Tennessee.

Oswald, S.E., Menon, M., Carminati, A., Vontobel, P., Lehmann, E., Schulin, R., 2008. Quantitative imaging of infiltration, root growth, and root water uptake via neutron radiography. Vadose Zone J. 7, 1035–1047. https://doi.org/10.2136/vzj2007.0156.

Pertassek, T., Peters, A., Durner, W., 2015. HYPROP-FIT software user's manual, V.3.0. In: AG, M.G. (Ed.), Mettlacher Str. 8, 81379 München, p. 66. Germ any.

Peters, A., Durner, W., 2008. Simplified evaporation method for determining soil hydraulic properties. J. Hydrol. 356, 147–162. https://doi.org/10.1016/j.jhydrol.2008.04.016.

Peters, A., Durner, W., 2015. SHYPFIT 2.0 User's Manual Institut für Ökologie. Technische Universität Berlin, Germany.

Peters, A., Iden, S.C., Durner, W., 2015. Revisiting the simplified evaporation method: identification of hydraulic functions considering vapor, film and corner flow. J. Hydrol. 527, 531–542. https://doi.org/10.1016/j.jhydrol.2015.05.020.

Petersson, H., Messing, I., Steen, E., 1987. Influence of root mass on saturated hydraulic conductivity in arid soils of central Tunisia. Arid Soil Res. Rehabil. 1, 149–160. https://doi.org/10.1080/15324988709381140.

Powis, V.B., Whalley, W.R., Bird, N.R.A., Leeds-Harrison, P.B., 2003. Some mechanisms responsible for the alteration of soil hydraulic properties by root activity, 16. In: Moore, H.M., et al. (Eds.), Land Reclamation-Extending Boundaries: Proceedings of

the 7th International Conference, vol. 13. May Swets and Zeitlinger, Lisse, Runcorn, UK, p. 404.
Querejeta, J.I., 2017. Soil water retention and availability as influenced by mycorrhizal symbiosis. In: Johnson, N.C., et al. (Eds.), Mycorrhizal Mediation of Soil: Fertility Structure and Carbon Storage Elsevier, pp. 299–317.
Rachman, A., Anderson, S.H., Gantzer, C.J., Alberts, E.E., 2004. Soil hydraulic properties influenced by stiff-stemmed grass hedge systems. Soil Sci. Soc. Am. J. 68, 1386–1393. https://doi.org/10.2136/sssaj2004.1386.
Rajamanthri, K., Jotisankasa, A., Aramrak, S., 2021. Effects of *Chrysopogon zizanioides* root biomass and plant age on hydro-mechanical behavior of root-permeated soils. Int. J. Geosynth. Ground Eng. 7 (36), 1–13. https://doi.org/10.1007/s40891-021-00271-0.
Ray, P., Guo, Y., Chi, M., Krom, N., Boschiero, C., Watson, B., Huhman, H., Zhao, P., Singan, V., Lindquist, E., Yan, J., Adam, C., Craven, K., 2021. *Serendipita* fungi modulate the switchgrass root transcriptome to circumvent host defenses and establish a symbiotic relationship. Mol. Plant Microbe Interact. 34 (10), 1128–1142.
Robinson, D.A., Hopmans, J.W., Filipovic, V., Van Der Ploeg, M., Lebron, I., Jones, S.B., Reinsch, S., Jarvis, N., Tuller, M., 2019. Global environmental changes impact soil hydraulic functions through biophysical feedbacks. Global Change Biol. 25, 1895–1904. https://doi.org/10.1111/gcb.14626.
Samaei, F., Asghari, S., Aliasgharzad, N., 2015. The effects of two arbuscular mycorrhizal fungi on some physical properties of a sandy loam soil and nutrients uptake by spring barley. J. Soil Environ. 1, 1–9.
SAS Institute Inc, 2012. SAS/STAT Version 9.4. Computer Software, Cary, NC, USA.
Scanlan, C., 2009. Processes and Effects of Root Induced Changes to Soil Hydraulic Properties School of Earth and Environment. University of Western Australia.
Scanlan, C., Hinz, C., 2010. Insight into the processes and effects of root induced changes to soil hydraulic properties. In: Proceedings of the 19th World Congress of Soil Science. Soil Solutions For a Changing World.
Schindler, U., Durner, W., Von Unold, G., Müller, L., 2010. Evaporation method for measuring unsaturated hydraulic properties of soils: extending the measurement range. Soil Sci. Soc. Am. J. 74, 1071–1083. https://doi.org/10.2136/sssaj2008.0358.
Scholl, P., Leitner, D., Kammerer, G., Loiskandl, W., Kaul, H.P., Bodner, G., 2014. Root induced changes of effective 1D hydraulic properties in a soil column. Plant Soil 381, 193–213. https://doi.org/10.1007/s11104-014-2121-x.
Scott, H.D., 2000. Soil Physics: Agricultural and Environmental Applications, first ed. Wiley-Blackwell, Iowa State University Press, Ames, IA.
Sedgley, R.H., Barley, K.P., 1954. Effects of root growth and decay on the permeability of a synthetic sandy loam. Soil Sci. 78, 205–210.
Sedgley, R.H., Barley, K.P., 1958. Effects of root growth and decay on the capillary conductivity of a sandy loam at low soil-moisture tension. Soil Sci. 86, 175–179. https://doi.org/10.1097/00010694-195810000-00002.
Seneviratne, S.I., Corti, T., Davin, E.L., Hirschi, M., Jaeger, E.B., Lehner, I., Orlowsky, B., Teuling, A.J., 2010. Investigating soil moisture–climate interactions in a changing climate: a review. Earth Sci. Rev. 99, 125–161.
Shao, W., Ni, J., Leung, A.K., Su, Y., Ng, C.W.W., 2017. Analysis of plant root–induced preferential flow and pore-water pressure variation by a dual-permeability model. Can. Geotech. J. 54, 1537–1552. https://doi.org/10.1139/cgj-2016-0629.
Thomas, R.S., Dakessian, S., Ames, R.N., Brown, M.S., Bethlenfalvay, G.J., 1986. Aggregation of a silty clay loam soil by mycorrhizal onion roots. Soil Sci. Soc. Am. J. 50, 1494–1499. https://doi.org/10.2136/sssaj1986.03615995005000060023x.
US Silica Company, 2010. Material Safety Data Sheet-Silica Sand and Ground Silica. US Silica Company.
Van Der Heijden, M.G.A., Streitwolf-Engel, R., Riedl, R., Siegrist, S., Neudecker, A., Ineichen, K., Boller, T., Wiemken, A., Sanders, I.R., 2006. The mycorrhizal contribution to plant productivity, plant nutrition and soil structure in experimental grassland. New Phytol. 172, 739–752. https://doi.org/10.1111/j.1469-8137.2006.01862.x.
van Genuchten, M.T., 1980. A closed-form equation for predicting the hydraulic conductivity of unsaturated Soils1. Soil Sci. Soc. Am. J. 44, 892–898. https://doi.org/10.2136/sssaj1980.03615995004400050002x.
Verma, S., Varma, A., Rexer, K.-H., Hassel, A., Kost, G., Sarbhoy, A., Bisen, P., Bütehorn, B., Franken, P., 1998. *Piriformospora indica*, gen. et sp. nov., a new root-colonizing fungus. Mycologia 90, 896–903. https://doi.org/10.2307/3761331.
Warren, J.M., Brooks, J.R., Meinzer, F.C., Eberhart, J.L., 2008. Hydraulic redistribution of water from *Pinus ponderosa* trees to seedlings: evidence for an ectomycorrhizal pathway. New Phytol. 178, 382–394.
Weiß, M., Waller, F., Zuccaro, A., Selosse, M.A., 2016. Sebacinales – one thousand and one interactions with land plants. New Phytol. 211, 20–40. https://doi.org/10.1111/nph.13977.
Whalley, W.R., Riseley, B., Leeds-Harrison, P.B., Bird, N.R.A., Leech, P.K., Adderley, W. P., 2005. Structural differences between bulk and rhizosphere soil. Eur. J. Soil Sci. 56, 353–360. https://doi.org/10.1111/j.1365-2389.2004.00670.x.
Yan, W.M., Zhang, G., 2015. Soil-water characteristics of compacted sandy and cemented soils with and without vegetation. Can. Geotech. J. 52, 1331–1344. https://doi.org/10.1139/cgj-2014-0334.
Zhuang, J., Flury, M., Jin, Y., 2003. Colloid-facilitated Cs transport through water-saturated Hanford sediment and Ottawa sand. Environ. Sci. Technol. 37, 4905–4911. https://doi.org/10.1021/es0264504.